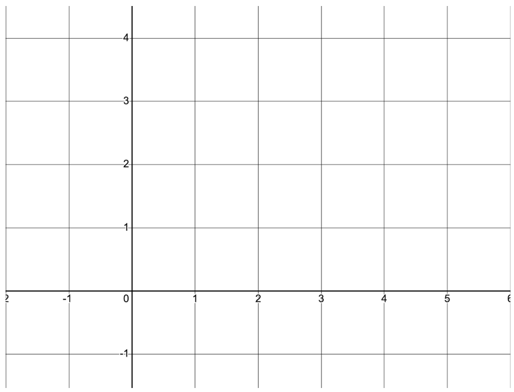
1. Consider the region bounded by $y = \sqrt{x}$, x-axis and $x = 4$.

1). Sketch the graph of the above region, and make sure to shade the region.



2). Set up integrals to compute the volume of the solid of revolution obtained by rotating the region in Part 1) about the y-axis using both disk/washer and cylindrical shell methods.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\_\_\_}^{\_\_\_} \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ \, d\_\_\_$$

ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\_\_\_}^{\_\_\_} \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ \, d\_\_\_$$

3). Use either one of the integrals in Part 2) to compute the volume of the solid obtained by rotating the given region about y-axis. Simplify the answer.